The probabilities are

$$f(0) = P(X=0) = \frac{25}{36},$$

$$f(1) = P(X=1) = \frac{10}{36}$$

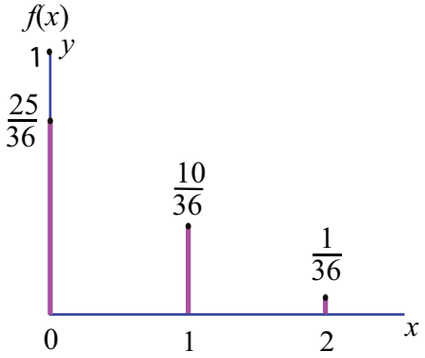
and $$f(2) = P(X=2) = \frac{1}{36}$$

Clearly the function $f(x)$ satisfies the conditions

(i) $f(x) \geq 0$, for $x = 0, 1, 2$ and

(ii) $$\sum_x f(x) = \sum_{x=0}^{x=2} f(x) = f(0) + f(1) + f(2) = 1$$

$$= \frac{25}{36} + \frac{10}{36} + \frac{1}{36} = 1$$



Probability mass function of $f(x)$

Fig. 11.5

The probability mass function is presented as

x	0	1	2
$f(x)$	$\frac{25}{36}$	$\frac{10}{36}$	$\frac{1}{36}$

(or) $$f(x) = \begin{cases} \frac{25}{36} & \text{for } x = 0 \\ \frac{10}{36} & \text{for } x = 1 \\ \frac{1}{36} & \text{for } x = 2 \end{cases}$$

11.3.3 Cumulative Distribution Function or Distribution Function

There are many situations to compute the probability that the observed value of a random variable X will be less than or equal to some real number x. Writing $F(x) = P(X \leq x)$ for every real number x, we call $F(x)$, the cumulative distribution function or distribution function of the random variable X and its common abbreviation is cdf.

Definition 11.4: (cumulative distribution function)

The **cumulative distribution function** $F(x)$ of a discrete random variable X, taking the values $x_1, x_2, x_3, \ldots$ such that $x_1 < x_2 < x_3 < \cdots$ with probability mass function $f(x_i)$ is

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i), \qquad x \in \mathbb{R}$$

The distribution function of a discrete random variable is known as Discrete Distribution Function. Although, the probability mass function $f(x)$ is defined *only* for a set of discrete values $x_1, x_2, x_3, \ldots$, the cumulative distribution function $F(x)$ is defined for all real values of $x \in \mathbb{R}$.

We can compute the cumulative distribution function using the probability mass function

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i) = \sum_{x_i \leq x} P(X = x_i)$$

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